NEWTON SCHOOL

PRE-CALCULUS | ANSWER KEY

Test Prep Answer Key — Induction · Recursion · Series · Bernoulli

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Companion answer key to Test Prep Worksheet — 12 Problems (no answers) · 1-to-1 mapped.

Read after you've attempted

Each solution shows the full **Base · IH · Step** skeleton for induction problems and the **setup · simplification · final** chain for the rest. Final answers are boxed in green. Cross-check against your worksheet work.

Topic 1 — Mathematical Induction

Problem 1 Sum of cubes — $\sum k^3 = [n(n+1)/2]^2$

Induction | Sum of cubes

STATEMENT

Prove by induction that $1^3 + 2^3 + 3^3 + \cdots + n^3 = \left[\frac{n(n+1)}{2}\right]^2$ for all $n \geq 1$.

SOLUTION

Base ($n = 1$). LHS $= 1^3 = 1$. RHS $= [1 \cdot 2/2]^2 = 1^2 = 1$. ✓

IH. $1^3 + \cdots + k^3 = [k(k+1)/2]^2$.

Step. Add $(k+1)^3$:

$$\sum_{i=1}^{k+1} i^3 = \left[\frac{k(k+1)}{2}\right]^2 + (k+1)^3.$$

Factor $(k+1)^2$:

$$= (k+1)^2\left[\frac{k^2}{4} + (k+1)\right] = (k+1)^2 \cdot \frac{k^2+4k+4}{4} = \frac{(k+1)^2(k+2)^2}{4} = \left[\frac{(k+1)(k+2)}{2}\right]^2. \checkmark$$

Conclusion. True for all $n \geq 1$. ■

Problem 2 $13^n - 5^n$ is divisible by 8

Induction | Divisibility

STATEMENT

Prove by induction that $13^n - 5^n$ is divisible by 8 for all $n \geq 1$.

SOLUTION

Base ($n = 1$). $13 - 5 = 8 = 8 \cdot 1$. ✓

IH. $13^k - 5^k = 8m$ for some integer m.

Step. Rewrite $13^{k+1} - 5^{k+1}$ by injecting a $-13 \cdot 5^k + 13 \cdot 5^k$ term:

$$13^{k+1} - 5^{k+1} = 13(13^k - 5^k) + 5^k(13 - 5) = 13(8m) + 8 \cdot 5^k = 8\,(13m + 5^k). \checkmark$$

Both terms are multiples of 8.

Conclusion. $8 \mid 13^n - 5^n$ for all $n \geq 1$. ■

Problem 3 Every $n \geq 2$ = product of primes (strong induction)

Strong Induction | Number Theory

STATEMENT

Prove by **strong induction** that every integer $n \geq 2$ can be written as a product of (one or more) prime numbers.

SOLUTION

Base ($n = 2$). 2 is prime — trivially a "product of one prime." ✓

Strong IH. Every integer j with $2 \leq j \leq k$ is expressible as a product of primes.

Step ($n = k + 1$). Two cases:

- If $k + 1$ is *prime*: it is itself a product of one prime. ✓
- If $k + 1$ is *composite*: write $k + 1 = a \cdot b$ with $2 \leq a, b \leq k$. By strong IH, both a and b are products of primes. Concatenate those factorizations → $k + 1$ is a product of primes. ✓

Conclusion. Every integer $n \geq 2$ is expressible as a product of primes. ■

Insight. This is the **existence** half of the Fundamental Theorem of Arithmetic. (Uniqueness needs a separate argument.)

Topic 2 — Run-of-Heads · Recursion

Problem 4 No-run-of-3-heads recurrence + first values

Recurrence | Probability

STATEMENT

Let $p(n)$ = probability that in n fair coin flips there is no run of 3 heads in a row. Set up the recurrence by splitting on how the sequence begins (T, HT, HHT) and state $p(1), p(2), p(3)$.

SOLUTION

Case split on the prefix. Each branch must avoid HHH:

- Start with **T** (prob $\frac{1}{2}$) → remaining $n-1$ flips need no run of 3.
- Start with **HT** (prob $\frac{1}{4}$) → remaining $n-2$ flips need no run of 3.
- Start with **HHT** (prob $\frac{1}{8}$) → remaining $n-3$ flips need no run of 3.
- Start with **HHH**: already a run of 3 — excluded.

Recurrence:

$$\boxed{p(n) = \tfrac{1}{2}\,p(n-1) + \tfrac{1}{4}\,p(n-2) + \tfrac{1}{8}\,p(n-3)} \quad (n \geq 3).$$

Initial values. $p(0) = 1$ (vacuous), $p(1) = 1$, $p(2) = 1$ (no way to fit HHH yet), $p(3) = 1 - \frac{1}{8} = \frac{7}{8}$ (only HHH violates).

$$\boxed{p(1) = 1,\ \ p(2) = 1,\ \ p(3) = \tfrac{7}{8}}$$

Problem 5 Compute $p(5)$ as a fraction in lowest terms

Recurrence | Probability

STATEMENT

Using the recurrence from #4, compute $p(5)$ as a fraction in lowest terms.

SOLUTION

Use $p(n) = \frac{1}{2}p(n-1) + \frac{1}{4}p(n-2) + \frac{1}{8}p(n-3)$.

$p(4)$: $\frac{1}{2}\cdot p(3) + \frac{1}{4}\cdot p(2) + \frac{1}{8}\cdot p(1) = \frac{1}{2}\cdot\frac{7}{8} + \frac{1}{4}\cdot 1 + \frac{1}{8}\cdot 1 = \frac{7}{16} + \frac{4}{16} + \frac{2}{16} = \frac{13}{16}$.

$p(5)$: $\frac{1}{2}\cdot p(4) + \frac{1}{4}\cdot p(3) + \frac{1}{8}\cdot p(2) = \frac{1}{2}\cdot\frac{13}{16} + \frac{1}{4}\cdot\frac{7}{8} + \frac{1}{8}\cdot 1$
$= \frac{13}{32} + \frac{7}{32} + \frac{4}{32} = \frac{24}{32} = \frac{3}{4}$.

$$\boxed{p(5) = \frac{3}{4}}$$

Sanity check. Among 32 length-5 sequences, **24** avoid HHH (those with HHH form the 8 sequences HHH??, ?HHH?, ??HHH minus overlaps = 8 — confirms $32 - 8 = 24$).

Problem 6

Binary strings of length n with no two consecutive 1's — find $a(6)$

Recurrence | Fibonacci

STATEMENT

Let $a(n)$ = number of binary strings of length n with no two consecutive 1's. Write the recurrence (with base cases) and compute $a(6)$.

SOLUTION

Case split on the last digit:

- Last digit 0 → first $n-1$ digits free of "11" → $a(n-1)$ options.
- Last digit 1 → second-to-last must be 0 → first $n-2$ digits free of "11" → $a(n-2)$ options.

Recurrence: $a(n) = a(n-1) + a(n-2)$, with $a(1) = 2$ (0, 1) and $a(2) = 3$ (00, 01, 10).

Compute up to 6: $a(3) = 5,\ a(4) = 8,\ a(5) = 13,\ a(6) = 21.$

$$\boxed{a(6) = 21}$$

Fibonacci connection. $a(n) = F_{n+2}$. Here $a(6) = F_8 = 21$.

Topic 3 — Series · Σ · Domain of Convergence

Problem 7 Arithmetic sum $\sum_{k=1}^{20}(3k-2)$

Arithmetic Series

STATEMENT

Find the sum $\displaystyle\sum_{k=1}^{20}(3k-2)$.

SOLUTION

Linearity. $\displaystyle\sum_{k=1}^{20}(3k-2) = 3\sum_{k=1}^{20}k - \sum_{k=1}^{20}2 = 3\cdot\frac{20\cdot 21}{2} - 2\cdot 20.$

$= 3\cdot 210 - 40 = 630 - 40 = 590.$

$$\boxed{\sum_{k=1}^{20}(3k-2) = 590}$$

Problem 8 Geometric series — domain of convergence + closed form

Geometric Series | Domain of Convergence

STATEMENT

Determine the **domain of convergence** and the sum in closed form for
$5 + 5\left(\frac{x+1}{2}\right) + 5\left(\frac{x+1}{2}\right)^2 + \cdots$

SOLUTION

Identify. Geometric with first term $a = 5$ and common ratio $r = \dfrac{x+1}{2}$.

Convergence condition. $|r| < 1 \iff \left|\dfrac{x+1}{2}\right| < 1 \iff |x+1| < 2 \iff -3 < x < 1.$

Closed-form sum (on that domain).

$$S = \frac{a}{1-r} = \frac{5}{1-\frac{x+1}{2}} = \frac{5}{\frac{2-(x+1)}{2}} = \frac{5\cdot 2}{1-x} = \frac{10}{1-x}.$$

Domain: $-3 < x < 1$ · **Sum:** $\dfrac{10}{1-x}$

Problem 9 Geometric: $a = 6, r = 2/3 - S_\infty$ and the smallest n within 0.01

Geometric Series | Convergence Rate

STATEMENT

A geometric series has first term $a = 6$ and common ratio $r = \frac{2}{3}$. Find S_∞, and find the smallest n such that S_n is within 0.01 of S_∞.

SOLUTION

S_∞. $S_\infty = \dfrac{a}{1-r} = \dfrac{6}{1-2/3} = \dfrac{6}{1/3} = 18.$

Tail size. $S_n = a\dfrac{1-r^n}{1-r} = 18(1-(2/3)^n)$, so $|S_\infty - S_n| = 18\,(2/3)^n$.

Solve $18\,(2/3)^n < 0.01$. $(2/3)^n < \frac{1}{1800} \iff n > \dfrac{\ln 1800}{\ln(3/2)} \approx \dfrac{7.4955}{0.4055} \approx 18.49$. So $n = 19$.

Verify. $18\,(2/3)^{18} \approx 0.0122 > 0.01;\ \ 18\,(2/3)^{19} \approx 0.0081 < 0.01.$ ✓

$$\boxed{S_\infty = 18, \quad n_{\min} = 19}$$

Topic 4 — Bernoulli's Madness

Problem 10 Hockey stick: $\sum k(k+1)(k+2)$ closed form

Hockey Stick | Continuous Products

STATEMENT

Use the hockey stick identity to evaluate $\displaystyle\sum_{k=1}^{N} k(k+1)(k+2)$ in closed form.

SOLUTION

Hockey stick form. $\displaystyle\sum_{k=0}^{N} k(k+1)\cdots(k+j-1) = j!\binom{N+j}{j+1}$. Here $j = 3$.

$$\sum_{k=1}^{N} k(k+1)(k+2) = 3!\binom{N+3}{4} = 6\cdot\frac{(N+3)(N+2)(N+1)N}{24} = \frac{N(N+1)(N+2)(N+3)}{4}.$$

$$\boxed{\sum_{k=1}^{N} k(k+1)(k+2) = \frac{N(N+1)(N+2)(N+3)}{4}}$$

Memory trick. j continuous terms in the summand → answer is $(j+1)$ continuous terms divided by $(j+1)$.

Problem 11 Telescoping: $\sum 1/[k(k+1)]$

Telescoping | Partial Fractions

STATEMENT

Use telescoping to evaluate $\displaystyle\sum_{k=1}^{N} \frac{1}{k(k+1)}$.

SOLUTION

Partial fractions. $\displaystyle\frac{1}{k(k+1)} = \frac{1}{k} - \frac{1}{k+1}$.

Telescope.

$$\sum_{k=1}^{N}\left(\frac{1}{k} - \frac{1}{k+1}\right) = \left(1 - \tfrac{1}{2}\right) + \left(\tfrac{1}{2} - \tfrac{1}{3}\right) + \cdots + \left(\tfrac{1}{N} - \tfrac{1}{N+1}\right) = 1 - \frac{1}{N+1} = \frac{N}{N+1}.$$

$$\boxed{\sum_{k=1}^{N} \frac{1}{k(k+1)} = \frac{N}{N+1}}$$

Problem 12 Linear recurrence $T_n = 2T_{n-1} + 3,\ T_0 = 1$ — closed form + T_5

Linear Recurrence | Geometric Series

STATEMENT

Solve $T_n = 2T_{n-1} + 3$ with $T_0 = 1$. Find a closed form for T_n and compute T_5.

SOLUTION

Find a particular (steady) solution. Try constant $T_p = c$: then $c = 2c + 3 \Rightarrow c = -3$.

Homogeneous part. $T_h = A \cdot 2^n$.

General solution. $T_n = A \cdot 2^n - 3$.

Apply initial. $T_0 = 1 \Rightarrow A - 3 = 1 \Rightarrow A = 4$. So $T_n = 4 \cdot 2^n - 3 = 2^{n+2} - 3$.

Evaluate at $n = 5$. $T_5 = 2^7 - 3 = 128 - 3 = 125$.

$$\boxed{T_n = 2^{n+2} - 3, \quad T_5 = 125}$$

Verify chain. $T_0 = 1, T_1 = 5, T_2 = 13, T_3 = 29, T_4 = 61, T_5 = 125$. Each $T_{n+1} = 2T_n + 3$ — checks.

Answer Key paired with *Test Prep Worksheet — 12 Problems (no answers)* © Copyright Newton School.